

Design of a 13.56 MHz RFID system

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Abstract — This paper presents a 13.56 MHz RFID reader that can be used as a door-lock for home security. The RFID reader consists of a transmitter, a receiver, and a data processing block. To verify the operation of the developed RFID reader, we present both a PSPICE simulation for transmitter/receiver and a digital simulation for data processing block. In particular, a CRC block for error detection of received data and a Manchester decoding block for position detection of collided data are designed using VHDL. In addition, we applied a binary search algorithm to multi-tag anti-collision. The anti-collision procedure is carried out by PIC microcontroller on software. Both the experimental and simulation results show that the developed reader can provide the right multi-tag recognition.

Keywords — RFID, anti-collision, binary search algorithm.

1. Introduction

A RFID (Radio Frequency Identification) technology is a non-contact method for information identification technology using radio frequency. RFID is practical technology in industry due to continuous growth of the Internet, manufacturing technology of low-cost tag, and standardization of electrical identification code.

The RFID technology is originally developed in the U.S.A for military purpose of missile tracking. The RFID implies a radio frequency system which consists of tag (or transponder) including electronic microchip and reader (or interrogator) for data read and write. The data exchange between tag and reader is achieved by using magnetic or electromagnetic fields. Production data stored in the microchip can be transferred to several meters or more than ten meters by antenna. After decoding the tag data, reader sends them to the host computer. Because RFID has relatively long transmission range and multi-tag recognition, it will be a core technology which can replace conventional bar-code system used in the field requires safety, security, and logistics [1-5].

In this paper, we consider a system modeling process for design of RFID reader's transmitting section and

receiving section using PSPICE. In addition, CRC (Cyclic Redundancy Check) module for error detection and Manchester coding module for collision detection are designed by VHDL. Both PSPICE simulation and VHDL simulation results show the validity and verification of system design.

The remainder of this paper is organized as follows. Section 2 describes design and implementation of a 13.56 MHz RFID reader. Section 3 presents our experimental results, and Section 4 concludes this paper.

2. A 13.56 MHz RFID System

RFID system typically consists of mobile radio frequency (RF) tags including a low-functional microchip and a RF-tag reader including transmitter/receiver module and a host computer. Figure 1 shows the RFID system configuration. RF-tag sends data to a reader using the radio frequency when it comes into the recognition field. RF-tag has a unique ID during the manufacturing process. RFID reader receives both tag ID and tag data during the system operating.

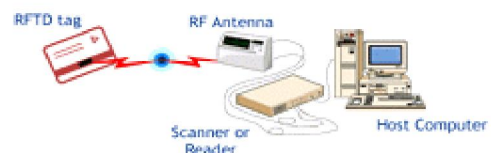


Figure 1. RFID system configuration diagram

Figure 2 shows the functional block diagram of a 13.56 MHz RFID reader. The RFID reader consists of three functional blocks. Transmitting block sends request commands to RF-tag in the recognition field as detailed in Section 2.1. Receiving block receives data from RF-tags through an antenna as detailed in Section 2.2. Data processing block deals with data as detailed in Section 2.3.

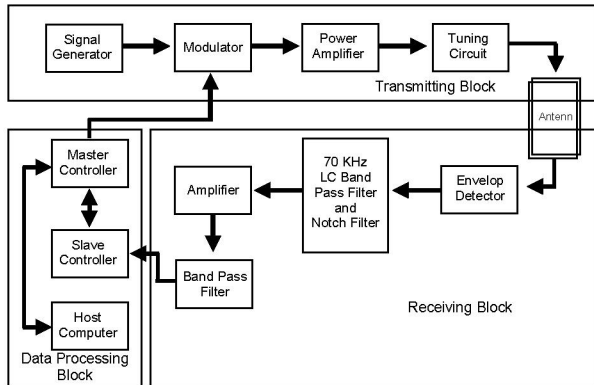


Figure 2. Block diagram of a 13.56MHz RFID reader

2.1 Transmitting block

Transmitting block contains a signal generator, a modulator, a power amplifier and a tuning circuit as shown in Figure 2. The signal generator generates carrier signal for RFID system. In this paper, we use a 13.56 MHz oscillator. The modulator plays an important role in transmitting block. It is classified by several blocks according to the kind of transmitters. The aim of the modulation is to increase noise immunity of the source signal and to reduce the size of transmission antenna. In order to design the modulator, we use a commercial device, NC7SZ08 from Fairchild Semiconductor, because high-speed operation is possible due to signal created in signal generator and control signal created in data processing part. The power amplifier does function that amplifies electric power of changing tone signal for transmitting signal having sufficient strength in transmitter end. The proposed power amplifier can deal with high dBm's electric power in output-end to drive large electric current unlike conventional amplifiers. And we can also control output level in electrical power by making use of voltage regulator that can variably control the electric power due to distance between tag and reader.

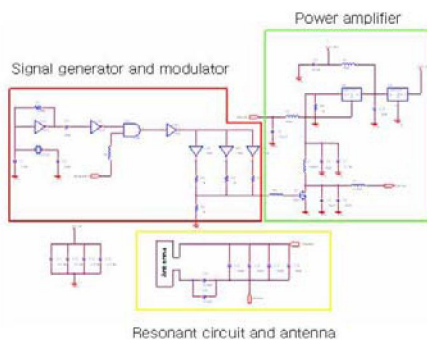


Figure 3. Transmitting block schematic

A resonant circuit can concentrate transmission of the message energy in any specific frequency. Because the resonant circuit has the selective properties for a

specific frequency, RFID system can chose only a particular frequency ingredient. Frequency selection occurs when a cycle of input signal and a unique period of resonator are matching each other. L-C resonance frequency is as follows.

$$f_r = \frac{1}{2\pi\sqrt{LC}} \quad (1)$$

Therefore, we can select a frequency with adjusting L value and C value properly. Figure 3 shows the detail schematic about the transmitting section.

2.2 Receiving block

Receiving block consists of detector, filter, and amplifier as shown in Figure 2. The envelop detector recovers received Manchester coded signal with a 70 KHz frequency rate from receiver antenna. It is designed with BAS21 device for high-speed switching operation. The proposed envelop detector has several advantages; it can handle received signal which is relatively high electric-power from the antenna, and there is no limitation of input signal, and a distortion decreases under increasing signal magnitude. LC band-pass filter is used to filter a received signal from tag. It is made of coil, capacitor and high-speed diode for the limitation of signal amplitude. Also, we adopt a notch filter to remove the noise due to carrier signal. Figure 4 shows the designed receiving block circuit.

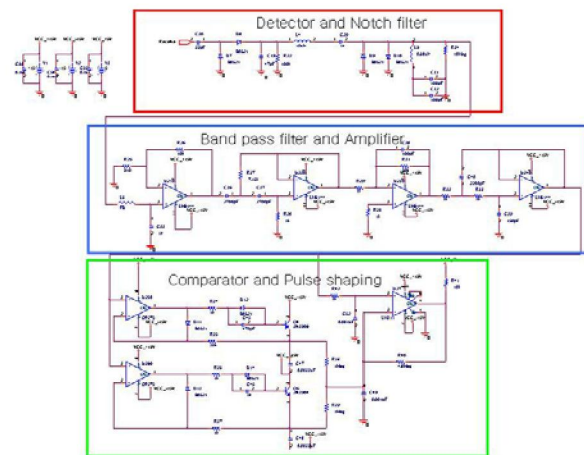


Figure 4. Receiving block schematic

In order to extract signal in a required band and amplify the signal in a specific voltage gain, we take advantage of multi-stage filters and band-pass filter. The band-pass filter is composed of a high-pass filter with 26dB gain and a low-pass filter with 29dB gain. Also, to achieve mean-value of signal from the band-pass filter, we use the commercial devices such as OP275, 2N3906, and 2N3904. Finally, we designed an extraction circuit of mean-value and a comparator for pulse shaping.

Down link (reader→tag)	REQUEST <11111111	1 st iteration	REQUEST <10111111	2 nd iteration
Uplink		1x1x001x		101x001x
Tag 1		10110010		10110010
Tag 2		10100011		10100011
Tag 3		10110011		10110011
Tag 4		11100011		

Down link (reader→tag)	REQUEST <10101111	3 rd iteration	SELECT 10100011	Read/Write
Uplink		10100011		
Tag 1				
Tag 2		10100011		10100011
Tag 3				
Tag 4				

Figure 5. Processing steps for collision-avoidance function

2.3 Data processing block

Two controllers are incorporated in data processing block. The first controller is targeted for decoding and communication of commands between reader and host computer. The second controller is used for collision avoidance, error detection, Manchester decoding, and CRC (Cyclic Redundancy Check) procedure creation for the coded transmission data. Manchester coding is used for data coding which interprets the coding data as '1' if the level of data bit has changed from high to low. On the other hand, it interprets the coding data as '0' if the level of data bit has changed from low to high. If there is no data change in the Manchester coding method, it means the collision of data. If we detect the part that is not to be changed, the exact position of the collision can be detected. For example, if we find out the correct position which collision occurs, the collision avoidance module sets the bit with '0' by binary search algorithm (Figure 5) and requires retransmission of a message from tag. The tag data without collision can be achieved if such process is carried out repeatedly.

During data exchange between RFID reader and tag, error occurs by interference. We made use of CRC-16/ITU module to detect this transmission error. A used generator for polynomial representation is $X^{16} + X^{12} + X^5 + 1$. An initial value of register is set to "FFFF". After receiving the first bit (LSB) in the first place, CRC-16 module does sequentially XOR arithmetic until the last bit (MSB) of data enters into it. The remainder of modulo-2 division, therefore, becomes the final CRC value. Figure 6 illustrates the scheme of CRC-16 modulo-2 division.

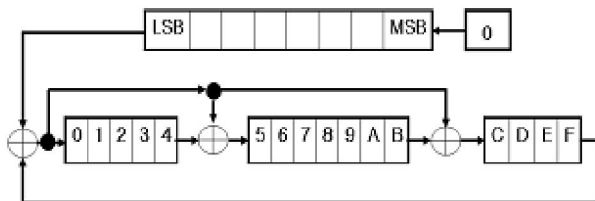


Figure 6. CRC 16/ITU generation scheme

3. Experimental results

Figure 7 shows the prototype using commercial elements which is implemented on PCB (Printed Circuit Board). The part in right side indicates the reader's main board, and the part in the middle is for data collision avoidance, and the part in left side is the antenna board, respectively.

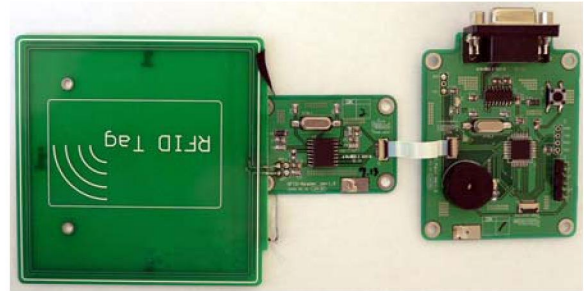


Figure 7. A 13.56 MHz RFID system

Figure 8 shows the measured waveforms of the transmitting and receiving signals through antenna; a base-band signal and a FFT waveform, respectively.

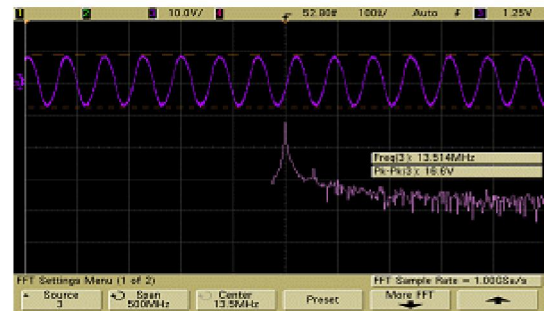


Figure 8. Measured antenna signals

Figure 9 illustrates the measured waveforms of the request signal from the reader and modulating signal altered by the request signal when tag is in recognition region. Figure 10 shows the recognized waveform when a single tag exists within recognition area.

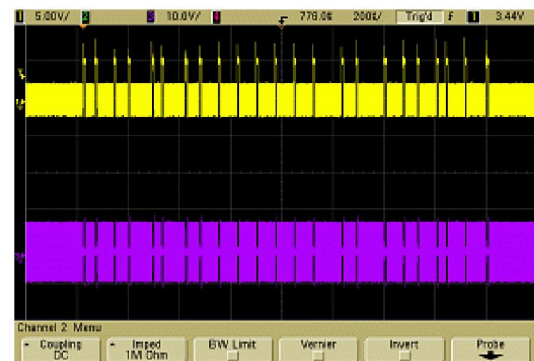


Figure 9. Request signal and modulated request signal

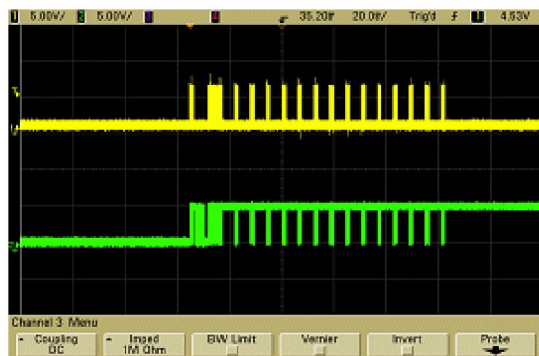


Figure 10. The measured waveforms with a single tag

Figure 11, 12 and 13 show the tag-recognition waveforms when two tags, four tags and five tags exist within recognition area.

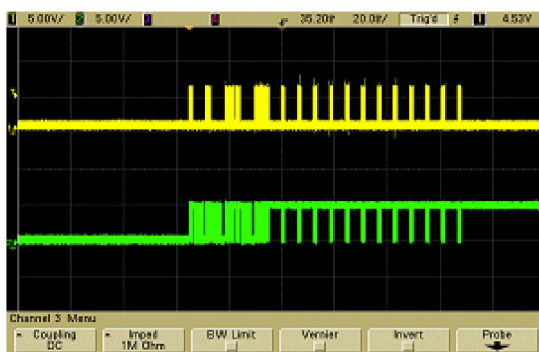


Figure 11. The measured waveforms with two tags

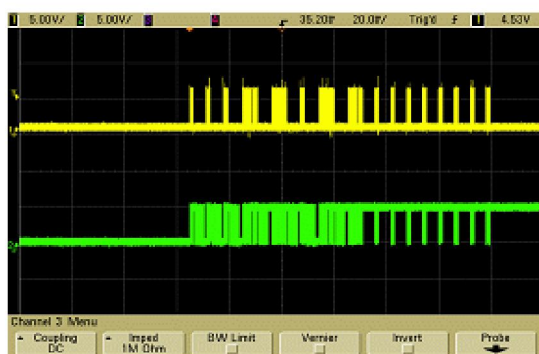


Figure 12. The measured waveforms with four tags

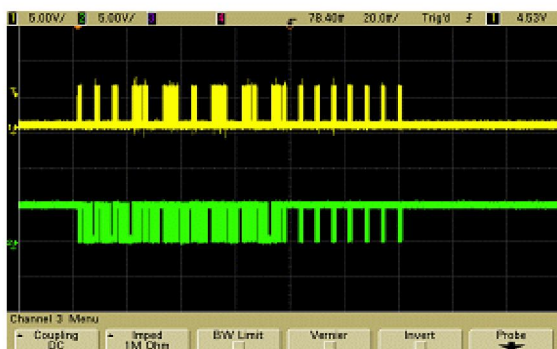


Figure 13. The measured waveforms with five tags

We can confirm that the number of recognized data-block has changed according to the number of tag. Moreover, measurement results show the exact awareness without collision under multi-tag environment.

4. Summary

A 13.56 MHz RFID system for multi-tag recognition has designed and implemented using commercial devices. The system consists of a transmitter/receiver module, a data processing module, and a RF tag. Transmitter is composed of sub-parts such as a signal generator, a power amplifier, and a resonant circuit. On the other hand, the receiver is composed of sub-parts such as an envelop detector, a notch filter, a band pass filter, a mean-value extractor, and a comparator. The data processing module consists of two processors; the one is master, the other is slave. The master processor takes charge of the whole control of system and the slave processor is for collision avoidance function using binary search algorithm. To verify the operation of the designed RFID reader, we show both a PSPICE simulation and VHDL simulation. The measured results demonstrate that the system recognize the multi-tag up to five.

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